

INDEPENDENT ANALYSIS

Novelty, Publishability & QACT Improvement Assessment

Autonomous Reinforcement Learning for Resonator Calibration (QRC / ARCS) — Software Scope Only

Prepared for	Akash Kishore — QRC / ARCS project (22ET49, DSCE / VTU)
Date	18 September 2026
Basis	QRC_ARCS_Project_Status_Report.docx (13 July 2026) + current literature search
Constraint honoured	Physical hardware cannot be built or tested. Only software, LTSpice, and optionally NI Multisim / MATLAB-Simulink circuit-level cross-checks are available.

1. Executive Verdict

Short answer: as currently framed, **no** — **this is not a novel contribution** to the reinforcement-learning-for-quantum-control field. It is a methodologically careful engineering exercise that re-runs, on a much smaller and entirely simulated scale, a question that has already been answered — repeatedly, on real superconducting qubit hardware, in top-tier venues, and in at least one filed patent — before this project began. That is not a reason to shelve it. It is a reason to stop framing it as “we show RL beats PID at frequency tracking” (already demonstrated, on real qubits) and start framing it as what it actually, defensibly is: a rigorous, reproducible, honestly-reported case study of RL control behaviour, failure modes, and cross-simulator transfer, built entirely from open tools. That second framing is publishable in an applied, technical-correctness-focused venue such as IEEE Access. The first framing is not, and will very likely draw a reviewer who knows this literature.

Two structural issues need to be fixed before anything else, independent of every other point in this document:

- **“Sim-to-real” is being used to describe a transfer that is actually sim-to-sim.** The status report calls QACT “a sim-to-real transfer data point.” It is not — it is a transfer between two simulators (a classical Gymnasium environment and Qiskit Aer). No physical hardware is involved anywhere in that experiment. “Sim-to-real” has a specific, well-understood meaning in the RL literature (simulation → physical device), and using it for a simulator-to-simulator handoff will read, to any reviewer who works in this area, as either a misunderstanding of the term or an attempt to inflate the result. Every instance of “sim-to-real” that does not involve real hardware needs to become “sim-to-sim” or “cross-simulator transfer.”
- **The paper's stated central contribution now has no path to completion.** Section 4.4 of the status report says the paper's “central contribution is the sim-to-real story (V4 simulation → QACT quantum-sim transfer → physical hardware transfer)” and Section 7 sequences everything around a five-phase hardware roadmap. You've now told me the physical build is not happening — not “delayed,” not “blocked on procurement,” but structurally out of scope. That means the narrative arc the existing report is written around cannot be finished. This has to be decided and rewritten now, not discovered by a reviewer later.

Everything below assumes the second point is correct — that hardware is permanently out of scope, and the paper must stand on simulation, cross-simulator transfer, and (optionally) higher-fidelity circuit-level co-simulation alone.

2. Novelty Analysis

2.1 What Already Exists

I searched current literature rather than relying only on prior knowledge, since this subfield moves fast and your own report is dated mid-2026. This is not an exhaustive review — it's enough to establish the bar you'd actually be measured against.

Work	Venue / Year	What it does	Overlap with QRC / ARCS
Fösel, Tighineanu, Weiss, Marquardt	Phys. Rev. X 8, 031084 (2018)	Foundational neural-network RL for quantum feedback control	Establishes RL-for-quantum-control as an ~8-year-old subfield, not a new one
Baum et al.	PRX Quantum 2, 040324 (2021)	Deep RL for error-robust gate-set design, trained and validated on real superconducting hardware	Real-hardware RL calibration, already done
Ding et al., two-qubit fluxonium gate paper	arXiv:2304.06087 (2023)	PPO (same algorithm as your V4) used to boost real CZ-gate calibration beyond a physics-based baseline, on real fluxonium/transmon hardware	Directly overlaps your planned Phase 4/5 — PPO + hardware calibration — already done, on

Work	Venue / Year	What it does	Overlap with QRC / ARCS
			real qubits, with PPO specifically
“Enhanced Qubit Readout via Reinforcement Learning”	Phys. Rev. Applied (2024/2025)	PPO agent trained in simulation, then tested live on real IBM “Kyoto” and “Brisbane” cloud devices; beats default calibration, 3× faster	The same train-in-sim-then-transfer pattern you attempt with QACT, but landed on real hardware with state-of-the-art results
“Towards Automated Superconducting Circuit Calibration using Deep RL (ASCC-DRL)”	IEEE conference (2021)	DRL calibration of a simulated transmon-cavity circuit model	Near-identical framing to your project title, already published five years ago
“Vibe Calibration”	arXiv:2606.22376 (2026)	LLM-agent-orchestrated, fully autonomous calibration of a real 112-qubit processor in 4.7 hours	Shows the field's current frontier is several methodological generations past a single-resonator PPO-vs-PID comparison
US Patent — automatic real-time calibration of qubit chips using RL	USPTO, granted	RL-based automatic real-time qubit calibration, claimed and granted	The core idea is not just published — it is patented

2.2 Is “RL beats PID at tracking” itself novel?

No — independent of quantum computing entirely. The same experimental pattern (train an RL agent, compare tracking error against a tuned PID baseline under some disturbance) turns up in piezo-actuator stage control, series-elastic-actuator force control, industrial/chemical process control, fixed-wing UAV attitude control under wind, aircraft trajectory tracking, and free-space laser-communication pointing control. Running this comparison once more on a new physical system — an LC resonator instead of an actuator or an aircraft — does not, by itself, constitute a novel contribution to either control theory or quantum computing. It's a standard, well-worn validation pattern, not a research finding.

2.3 Net Assessment

Combine 2.1 and 2.2: the algorithm (PPO), the baseline comparison (vs. PID), and the application domain (superconducting-qubit-adjacent frequency calibration) have each — individually and in combination — already been published, on real hardware, at higher fidelity, on larger and more capable systems, as recently as a few months before this conversation. Submitted as a novelty claim, this project will not survive review by anyone who knows this literature. Submitted as an honestly-scoped methodology/case-study paper, it has a real chance — that distinction is the whole point of this report.

3. What Is Actually Defensible, Given a Software-Only Constraint

Strip out the claims the literature above already owns, and three honest, narrower claims survive. None of them requires hardware. All three can be combined into one paper.

Candidate framing	What it actually claims	Strength as-is
A. Documented RL failure modes on a resonance-tracking task	MAE alone is an unreliable quality signal; a static, degenerate policy near resonance can score deceptively well. You've already caught this in V1, V4 OU-seed-2, and V4a, and built a trajectory-inspection safeguard against it.	Genuinely useful and under-reported in applied RL-control write-ups. Real but modest on its own — a methodology note, not a full paper.
B. Statistically rigorous negative/null result under OU drift	V4 does not significantly beat PID under Ornstein–Uhlenbeck drift, only under Gaussian drift. Reporting this cleanly, with effect size and a confidence interval rather than burying it, is good scientific practice that many applied papers skip.	Real, and reviewers at a technical-correctness-focused venue tend to respect honestly reported null results. Still modest alone.
C. Quantified, statistically characterized cross-simulator (classical → quantum-circuit-simulator) transfer	How much does a classically-trained control policy degrade when evaluated inside a different simulated physics stack, and can that degradation be explained (episode-level variance breakdown, noise injection, domain randomization)? This is the QACT work, correctly labeled and with the rigor upgrades in Section 5.	The most substantive of the three if done properly — it's a real (if small-scale) sim-to-sim robustness study, which is a recognized, if crowded, subfield.

Put together — A + B + C, explicitly framed as a reproducible, low-cost, fully-open methodology exercise rather than a scientific first — is a coherent, honest paper. It will not be read as a breakthrough, and it should not claim to be one. IEEE Access does not require a breakthrough (more on this in Section 7).

4. Improving the QACT / Qiskit Cross-Simulator Results — Software Only

You asked specifically for software-only improvements. Before the list, one clarification you need to resolve and state explicitly in the methods section, because a knowledgeable reviewer will ask:

Qiskit Aer's native PulseSimulator backend — the component that simulates continuous-time Hamiltonian dynamics under a driven pulse, which is what an amplitude-feedback, frequency-tracking control task like yours would need — was deprecated by the Qiskit team in 2023 in favour of the separate qiskit-dynamics library. If QACT is driving a continuous Rabi-type interaction (the `omega_rabi` parameter in your technical notes suggests it is) through Aer directly, you need to document exactly which simulation primitive is producing that dynamics: a discretized approximation built on Aer's statevector/density-matrix backends, an explicit Lindblad master-equation model, or qiskit-dynamics. “Qiskit Aer” alone is not a sufficient methods description for this task, and an unclear answer here is the single fastest way to lose a quantum-literate reviewer's confidence in the rest of the results.

With that resolved, here is what would materially strengthen the transfer result, in priority order:

Recommendation	Why it matters	Effort
Fix the label: “sim-to-sim”, not “sim-to-real”, everywhere	Terminology accuracy; see Section 1. This is not optional.	Trivial
Run QACT across multiple seeds, report mean ± CI, not a single ± SD from one run	A single run's std-dev (4581 Hz against a 3237 Hz mean) is exactly the kind of number that looks like noise from one unlucky run rather than a characterized distribution. You already do multi-seed reporting for V4 (Seed 3 chosen among others) — QACT	Medium — mainly compute time

Recommendation	Why it matters	Effort
	deserves the same rigor, especially since it's the headline transfer result.	
Break the variance down per-episode (you already proposed this — do it first)	If a subset of episodes (e.g. a certain drift phase, certain noise realization) drives most of the 4581 Hz spread, that's a real, specific, citable finding — “transfer fails under condition X” is a far stronger and more paper-worthy result than one aggregate number.	Low — post-hoc analysis of existing runs
Add a real Qiskit Aer noise model (NoiseModel / fake-backend), not an idealized simulator	Right now the “quantum” claim in QACT rests on it running inside Qiskit rather than on it modeling anything quantum-mechanically realistic (decoherence, readout error, gate error). Adding T1/T2 and readout noise via Aer's noise-model tooling makes the cross-simulator claim substantive rather than nominal.	Medium
Fine-tune, don't just zero-shot transfer	You're currently freezing V4 and dropping it into QACT cold. A small number of on-policy fine-tuning steps inside QACT (e.g. 5–10% of the original training budget) — reported as a separate, clearly-labeled experiment, not a change to locked V4 — quantifies adaptability, which is a stronger and more publishable result than raw zero-shot degradation. Report zero-shot and fine-tuned numbers side by side.	Medium–High
Re-tune the PID baseline natively inside QACT	Reusing V4's classical-environment PID gains inside a differently-scaled quantum-sim environment is not a fair baseline. A policy that “beats an untuned baseline” is a much weaker claim than one that beats a properly tuned one.	Low–Medium
Add observation/action noise injection and Q-factor domain randomization during classical (V5, future-work) training	Already proposed in your own Section 5.4 — this is correct and should be prioritized. Training on a distribution of plant dynamics rather than one fixed LC response is the standard sim-to-real/sim-to-sim robustness recipe and should directly narrow the QACT variance gap.	Medium–High
Verify VecNormalize statistics are ported correctly into the QACT observation scale	Already flagged in your Section 5.5 as a risk — a silent normalization mismatch between the classical env and QACT's observation space would fully explain an inflated variance without anything being wrong with the policy itself. Check this before attributing the variance to “real” transfer difficulty.	Low, but do it first — it could invalidate re-analysis done before it's fixed

One more software-only lever, and arguably the highest-value one available to you given the hardware constraint:

4.1 A “Virtual Hardware-in-the-Loop” Layer (Simulink / Multisim), Correctly Scoped

You mentioned NI Multisim and MATLAB-Simulink as available. Used correctly, these can do something genuinely useful that neither the abstracted Gymnasium environment nor QACT currently does: model the **non-idealities of the physical build** — 12-bit DAC/ADC quantization on the MCP4921/MCP3202, SPI transfer timing, OPA2134 gain-bandwidth and slew-rate limits, and realistic noise floors — as a higher-fidelity simulated plant. Re-running the frozen V4 policy (and, separately, a QACT-style transferred policy) against this more realistic virtual plant, instead of the idealized Gym environment, gives you a genuine intermediate data point between “pure abstract simulation” and “real hardware” — without needing real hardware.

Be precise about what this is and isn't in the paper: it is a higher-fidelity simulation, not hardware validation. Calling it “hardware-in-the-loop” without the word “virtual” or “simulated” in front of it recreates exactly the sim-to-real mislabeling problem from Section 1. Framed honestly (“virtual HIL” / “simulated plant with component-level non-idealities”), it is a legitimate and fairly novel-for-this-paper software-only contribution, and it's the closest defensible substitute you have for the physical validation you can't perform.

5. What Multisim / Simulink Can and Can't Do For This Paper

Use	Verdict
Cross-validate Stage 1/3/4 analog results against LTSpice (e.g. does an independent SPICE engine agree the resonator peaks at ~ 555.6 kHz, the AMS1117 rail holds ~ 3.3 V?)	Worth doing. Cheap, strengthens Section 3.3's credibility through tool-independent agreement, and directly resolves the open Stage 4 op-amp wiring caveat without hardware.
Build a control-theoretic transfer-function / state-space model of the resonator + PID for an analytical cross-check of the PID baseline	Worth doing. A short appendix deriving the PID baseline analytically (rather than purely empirically tuned) adds real rigor and answers a common reviewer question: “was the PID baseline actually competitive?”
Build the “virtual HIL” layer described in §4.1 (component non-idealities, quantization, SPI timing)	Worth doing — see §4.1. This is the strongest use of these tools for this paper.
Substitute for physical hardware validation, or support any “sim-to-real” claim about the RL policy, the QACT transfer, or the digital (Wokwi) stages	Cannot do this. Multisim and Simulink remain simulators. They do not touch real ADC/DAC quantization as fabricated, real wiring/RF parasitics, real SPI bus contention, or real component tolerance — which is the entire point a hardware stage would have served. Do not let any of this analysis be read as “a way around” the missing hardware; it strengthens Section 3 and adds a defensible §4.1 contribution, but it does not unlock the sim-to-real narrative in Sections 4.4 or 7 of the current report.

6. Publishability — A Straight Answer

6.1 There is no such thing as a 100%-probability paper

I want to be direct about this rather than answer around it: no framing, no amount of polish, and no venue choice produces a guaranteed acceptance. Peer review depends on which reviewers get assigned, their mood and bandwidth that week, what else is in the review pool competing for the same issue, and plain variance in human judgment — factors no author controls. Any guide that promises a formula for guaranteed acceptance is not being straight with you. What follows is how to maximize the probability, which is a different and honestly answerable question.

6.2 IEEE Access as a target, realistically

Current data on IEEE Access (checked 18 Sept 2026): it's a Scopus/SCIE/DOAJ-indexed, Q1–Q2, open-access mega-journal (impact factor figures I found range roughly 4.2–6.0 depending on source/year, publishing on the order of 13,000 articles/year), with commonly cited acceptance-rate figures in the 20–45% range across different sources — genuine peer review, not a rubber stamp, but explicitly scoped around technical correctness, methodological soundness, and clarity rather than field-defining novelty. That last point is important and works in your favour if the paper is reframed as Section 3 describes: IEEE Access does not require you to have out-novel'd Nature and PRX Quantum. It requires the work to be correct, complete, clearly presented, and honestly positioned against prior art.

6.3 What would currently get this desk-rejected or sent back for major revision

- Claiming or implying “sim-to-real” anywhere without real hardware in the loop (Section 1).
- A related-work section that omits the papers in Section 2.1 — a reviewer in this space will know at least some of them, and an absent or thin related-work section reads as either unfamiliarity with the field or an attempt to avoid the novelty question.
- Framing the contribution as “RL outperforms PID for resonance tracking” without immediately, explicitly positioning that against the real-hardware results already published (Section 2.1) — i.e., stating plainly that this is a small-scale, simulation-only replication built for accessibility/reproducibility, not a new finding about whether RL works for this class of problem.
- A methods section for QACT that says “Qiskit Aer” without specifying the actual dynamics model (Section 4).
- A single-run $\pm$ SD reported as if it were a characterized result, especially for the 4581 Hz QACT variance — reviewers who do RL will ask for multiple seeds.
- Any remaining Phase 1–5 hardware roadmap language presented as near-term future work, when it is not going to happen — this reads as scope creep or an unfulfilled promise once a reviewer checks the timeline against the rest of the paper.

6.4 A realistic, non-guaranteed estimate

If the sim-to-sim relabeling is done, the related-work section is honest and complete, the QACT methods are specified precisely, statistical rigor is added (multi-seed QACT, effect sizes for the OU null result), and the hardware narrative is replaced with the reframing in Section 3 — this becomes a plausible, defensible submission to IEEE Access, and a realistic informal estimate would be somewhere in the same 30–55% range the journal reports overall, possibly better than average given how much of the “technical correctness” bar (multi-tool cross-validation, careful statistics, transparent negative results) you're already hitting. That is a working estimate, not a promise — and it assumes the fixes below are actually made, not just planned.

7. Recommended Problem Statement & Paper Framing

7.1 Claims to make vs. claims to avoid

Avoid	Use instead
“We demonstrate that reinforcement learning can autonomously calibrate a resonant frequency, outperforming classical PID control.” (implies this is a new finding)	“We present a low-cost, fully reproducible, open-source classical-analog testbed and evaluation pipeline for studying RL-based resonance tracking, and use it to replicate and extend known RL-vs-PID findings under multiple stochastic drift regimes, including a statistically rigorous null result under correlated (OU) drift.”
“We show sim-to-real transfer from a classically-trained policy to a quantum-simulated environment.”	“We quantify zero-shot and fine-tuned cross-simulator transfer of a classically-trained PPO policy into a Qiskit-based quantum-circuit simulation, and characterize the sources of transfer variance.”
“This is the first system to...” / any first-of-its-kind language	“Unlike prior RL-for-qubit-calibration work, which requires access to real superconducting hardware [cite Fösel, Baum, Ding, et al.], this pipeline is reproducible on consumer-grade tools end-to-end, at the cost of not validating on physical qubits.” — name the trade-off, don't hide it.
Hardware Phases 1–5 as near-term roadmap / future work implying it's imminent	State plainly, once, in scope/limitations: physical hardware validation was out of scope for this study; results are simulation-only, cross-validated across LTSpice/Multisim/Simulink where applicable.

7.2 Suggested problem statement (draft)

A version you can adapt — deliberately modest and accurate rather than impressive-sounding:

“Reinforcement-learning-based calibration has been demonstrated on real superconducting qubit hardware, but that hardware is expensive, access-restricted, and impractical for iterative pedagogical or early-stage methodological study. This work builds an open, low-cost classical-circuit analog of the resonance-tracking problem central to qubit calibration — an LC resonator observed only through amplitude feedback, mirroring the indirect measurement constraints of dispersive qubit readout — and uses it to (1) rigorously compare a PPO-based control policy against a tuned PID baseline under both uncorrelated and correlated drift, including honest reporting of where the comparison is not statistically significant; (2) document and guard against a specific RL failure mode — degenerate near-resonance policies that score deceptively well on aggregate error metrics; and (3) quantify how much a classically-trained policy degrades when transferred, zero-shot and fine-tuned, into a Qiskit-based quantum-circuit simulation, as a lower-bound proxy for the sim-to-real gap that would arise on real hardware. All results are simulation-based and cross-validated across independent tools (LTSpice, [Multisim/Simulink]) where applicable; physical hardware validation is explicitly out of scope.”

7.3 Title options

- “A Low-Cost Classical Analog Testbed for Studying Reinforcement-Learning-Based Resonance Tracking Under Drift: Methodology, Failure Modes, and Cross-Simulator Transfer”
- “From PPO to Qiskit: Quantifying Cross-Simulator Transfer and Failure Modes in Reinforcement-Learning-Based Resonator Calibration”
- “Accessible Reinforcement Learning for Resonance Tracking: A Reproducible Classical-Analog Study with Cross-Simulator Validation”

8. Action Checklist

#	Action	Type
1	Decide and confirm: hardware is fully out of scope, not delayed. Rewrite Sections 4.4 and 7 of the status report around that decision before doing anything else.	Decision — do first
2	Global find-and-replace: “sim-to-real” → “sim-to-sim” / “cross-simulator transfer” everywhere it doesn't involve real hardware.	Writing fix
3	Write the related-work section against Section 2.1 of this report before writing anything else — it determines how every other claim has to be worded.	Writing
4	Document precisely what physics/dynamics primitive backs QACT (Aer statevector approximation vs. Lindblad model vs. qiskit-dynamics).	Methods clarification
5	Re-run QACT across multiple seeds; report mean $\pm$ CI. Do the per-episode variance breakdown you already proposed.	Experiment
6	Verify VecNormalize statistics parity between the classical env and QACT before trusting the current variance number.	Verification
7	Add a real Aer noise model (T1/T2, readout) to QACT; re-tune PID natively inside QACT as a fair baseline.	Experiment
8	Try fine-tuned (not just zero-shot) transfer into QACT as a separately labeled experiment; report both numbers.	Experiment, future-work-labeled
9	Build the LTSpice-vs-Multisim/Simulink cross-check for Stages 1/3/4 and resolve the Stage 4 wiring caveat analytically.	Verification
10	If time allows: build the “virtual HIL” layer in Simulink/Multisim (§4.1) — highest-value remaining software-only addition.	Stretch goal, high value
11	Rewrite the problem statement and title using §7 before drafting full sections.	Writing

Bottom line: the underlying engineering is solid and unusually well-instrumented for a mini-project — seed reporting, trajectory-inspection safeguards, honest null results, a documented degenerate-policy failure mode. None of that is wasted. What's missing is honest positioning against a field that has moved faster and further than the current draft acknowledges, and a paper narrative that no longer assumes hardware that isn't coming. Fix those two things and this becomes a publishable, if modest, technical paper. Leave them as-is and the most likely outcome is a reviewer who knows the literature catching both within the first two pages.